

Knowing Value Logic with Background Theories

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Background and Motivation

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- In his seminal work in 1989, Plaza proposed the public announcement logic. In the same paper, Plaza also introduced the knowing-value operator Kv, and listed a few valid formulas of $\mathbf{PAL} + Kv$. And the Kv operator is used to analyze the Sum-and-Product puzzle.
- Wang and Fan (2013) introduced the conditional knowing value operator $Kv_i(\varphi, a)$, and they provided a axiomatization for the resulting logic $\mathbf{ELKv}^r$ and also $\mathbf{PALKv}^r$ by reduction. The completeness proof of the multi-agent case was given in 2014.

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$$!\mathrm{Kv}^r : [\phi]Kv_i(\psi, a) \leftrightarrow \big(\phi \rightarrow Kv_i\big(\phi \wedge [\phi]\psi, a\big)\big)$$

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- Alexandru Baltag and collaborators extended knowing value logic with equality, function symbols, group knowledge, and theories of linear orders (2016, 2018, 2026).

Knowing Value Logic

Operator	Meaning	Logical Form	Example
$Kv_i a$	Agent i knows the value of a.	$\exists x\, K_i(a = x)$	"i knows what the password is."
$Kv_i(\phi, a)$	Agent i knows the value of a under condition ϕ.	$\exists x\, K_i(\phi \to a = x)$	"i knows the password, given that it is a 4-digit number."

Motivation

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- Standard knowing value logic can only express knowing the value of a non-rigid name. We aim to generalize it to include terms generated by function symbols.
- Knowing value logic has a close and natural connection with mathematical theories such as arithmetic. This motivates us to combine knowing value logic with background mathematical theories.

Sum and Product (van Ditmarsch and Kooi 2015)

The following is the Sum-and-Product puzzle. There are two integers x and y with $1 < x < y$ and $x + y \leq 100$. P knows the value of $x \times y$, and S knows the value of $x + y$. They have the following conversation:

- P: "I don't know the two numbers."
- S: "I knew that you didn't know. I don't know either."
- P: "Now I know the two numbers."
- S: "Now I know them too."

Consecutive Numbers (van Ditmarsch and Kooi 2015)

Anne and Bill get to hear the following: “Given are two natural numbers. They are consecutive numbers. I am going to whisper one of these numbers to Anne and the other number to Bill.” This happens. Anne and Bill now have the following conversation.

- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”
- Bill: “I know your number.”

How is that possible? What surely is one of the two numbers?

Formalization

To formalize the puzzle, the following statements are crucial:

- Both Anne and Bill knows that their numbers are consecutive.
- If Anne's number is 0, she knows Bill's number.

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To formalize the puzzle, the following statements are crucial:

- Both Anne and Bill knows that their numbers are consecutive.
- If Anne's number is 0, she knows Bill's number.

To formalize this puzzle, we are naturally led to:

- Introduce the successor function symbol S, the equality symbol $\approx$ and the constant symbol 0.
- Adopt a background theory of $(\mathbb{N}, \mathbf{S}, \mathbf{0})$, including basic axioms such as "0 is not a successor".

Knowing Value Logic with Successor Arithmetic

Language

Definition (Language PALKvSA^r(I, N))

Given a countable non-rigid name set **N** and a countable set of agents **I**, the set of terms Term and formulas are defined as follows.

$$t ::= 0 \mid a \mid St$$
$$\varphi ::= \top \mid t \approx t \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid Kv_i(\varphi, t) \mid [\varphi]\varphi$$

We write $Kv_i t$ as an abbreviation of $Kv_i(\top, t)$. And $\widehat{K}_i\varphi$ is an abbreviation of $\neg K\neg\varphi$.

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The announcement-free fragment is **ELKvSA**^{r}.

Semantics: Standard Models

Definition (Standard Models)

A (constant domain S5) standard model is a tuple $\mathcal{M} = \langle W, \{\sim_i\}_{i \in \mathsf{I}}, \mathfrak{N}, V_{\mathsf{N}} \rangle$, where $\sim_i$ is an equivalence relation on the set of possible worlds W for each agent i. Here $\mathfrak{N} = (\mathbb{N}, \mathsf{S}, \mathsf{0})$, and $V_{\mathsf{N}} : \mathsf{N} \times W \to \mathbb{N}$ assigns a natural number for each name-world pair.

Define the valuation for terms $V_{\mathrm{T}} : \mathrm{Term} \times W \to \mathbb{N}$ as follows:

$$V_{\mathrm{T}}(0, w) = \mathsf{0}$$
$$V_{\mathrm{T}}(a, w) = V_{\mathsf{N}}(a, w)$$
$$V_{\mathrm{T}}(St, w) = \mathsf{S}(V_{\mathrm{T}}(t, w))$$

Semantics

Then the satisfaction relation is defined as follows:

$$
\begin{array}{lcl}
\mathcal{M}, w \models t_1 \approx t_2 & \iff & V_{\mathrm{T}}(t_1, w) = V_{\mathrm{T}}(t_2, w) \\
\mathcal{M}, w \models Kv_i(\varphi, t) & \iff & \text{there exists an } n \in \mathbb{N}, \text{ for each } u \in W \\
 & & \text{such that } w \sim_i u \text{ and } \mathcal{M}, u \models \varphi, \\
 & & V_{\mathrm{T}}(t, u) = n. \\
\mathcal{M}, w \models [\varphi]\psi & \iff & \mathcal{M}, w \models \varphi \text{ implies } \mathcal{M}|_{\varphi}, w \models \psi
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\end{array}
$$

Proposition

ELKvSA^r *is not compact w.r.t. standard models.*

Proof.

The set $\{\neg a \approx S^n 0 \mid n \in \mathbb{N}\}$ is finitely satisfiable, but the whole set is not satisfiable. □

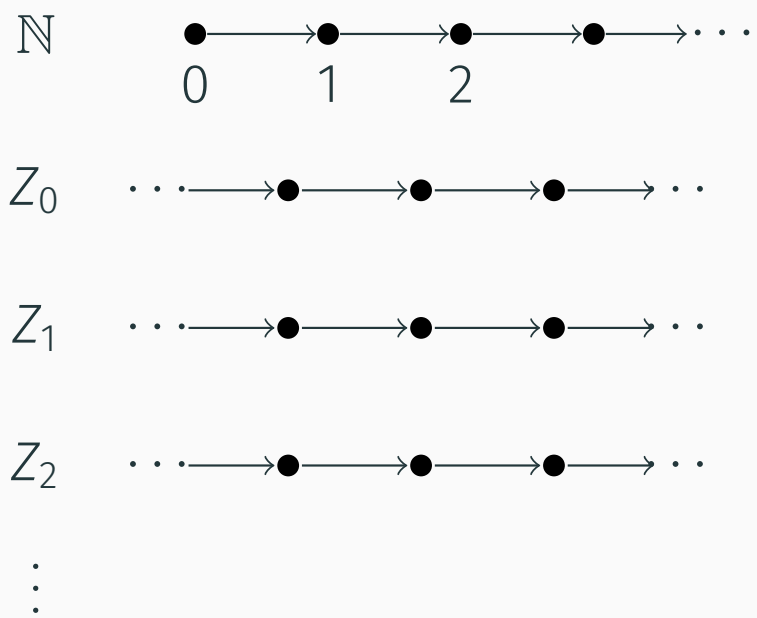
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We will define non-standard models, and prove the strong completeness w.r.t. non-standard models, and then use this result to prove the weak completeness w.r.t. standard models.

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We replace $(\mathbb{N}, \mathsf{S}, \mathsf{0})$ with the countably saturated model $(\mathbb{N}^*, \mathsf{S}^*, \mathsf{0})$.



Axiom System $\mathbb{ELKVSA}^r$

System $\mathbb{ELKVSA}^r$

Axiom Schemas

TAUT	all the instances of tautologies
DISTK	$K_i(\varphi \to \psi) \to (K_i\varphi \to K_i\psi)$
T	$K_i\varphi \to \varphi$
4	$K_i\varphi \to K_iK_i\varphi$
5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
ID	$t \approx t$
SI	$t \approx t' \to (\alpha \to \alpha(t'//t))$
	where α is an atomic formula
SA1	$\neg 0 \approx St$
SA2	$St_1 \approx St_2 \to t_1 \approx t_2$
SA3	$\neg S^n t \approx t$, where $n \geq 1$
Kv^r0	$Kv_i 0$
Kv^rS	$Kv_i(\varphi, t) \leftrightarrow Kv_i(\varphi, St)$
Kv^r4	$Kv_i(\varphi, t) \to K_i Kv_i(\varphi, t)$
KVE1	$Kv_i t_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$
KVE2	$Kv_i t_1 \wedge Kv_i t_2 \to (K_i(t_1 \approx t_2) \vee K_i\neg(t_1 \approx t_2))$

Rules

MP $\dfrac{\varphi \quad \varphi \to \psi}{\psi}$

NECK $\dfrac{\varphi}{K_i\varphi}$

RE $\dfrac{\psi \leftrightarrow \chi}{\varphi \leftrightarrow \varphi[\psi/\chi]}$

Kv^r-Elim $\dfrac{(Kv_i a \wedge K_i(\varphi \to a \approx t)) \to \psi}{Kv_i(\varphi, t) \to \psi}$

where a does not occur in φ, ψ, t

Strategy

To prove the weak completeness of $\mathbb{ELKVSA}^r$ w.r.t. standard models, we have the following strategy.

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- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models. But we don't use the Henkin-style construction.
- Then we show the finite world property, which is: if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.

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- Then we show the finite world property, which is: if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.
- Finally, we show that if ϕ is satisfiable in a finite non-standard model, then we can transform it into a finite standard model.

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- Then we show the finite world property, which is: if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.
- Finally, we show that if ϕ is satisfiable in a finite non-standard model, then we can transform it into a finite standard model.
- Thus, we can conclude that $\mathbb{ELKVSA}^r$ is weakly complete w.r.t. standard models. Moreover, **ELKvSA**^{r} is decidable.

Knowing Value Logic with Background Theories

Language

Definition (Language PALKVR$_\tau$)

Given a countably infinite sets $\mathbf{N}$ of non-rigid names, a set $\mathbf{I}$ of agents, and an at most countable first-order signature $\tau = \langle \mathbf{C}, \mathbf{F}, \mathbf{P} \rangle$ of rigid constant, function, and predicate symbols.

$$t ::= a \mid c \mid f(\bar{t})$$
$$\varphi ::= \top \mid t \approx t \mid P\bar{t} \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid Kv_i(\varphi, t) \mid [\varphi]\varphi$$

where $a \in \mathbf{N}$, $c \in \mathbf{C}$, $f \in \mathbf{F}$, $P \in \mathbf{P}$, and $i \in \mathbf{I}$. Function and predicate symbols have positive arity. We write $Kv_i t$ for $Kv_i(\top, t)$.

Language

Definition (Language PALKVR_τ)

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where $a \in \mathbf{N}$, $c \in \mathbf{C}$, $f \in \mathbf{F}$, $P \in \mathbf{P}$, and $i \in \mathbf{I}$. Function and predicate symbols have positive arity. We write $Kv_i t$ for $Kv_i(\top, t)$.

The announcement-free fragment is $\mathbf{ELKVR}_\tau$.

Semantics

Definition (Models)

A (constant domain S5) model is a tuple $\mathcal{M} = \langle W, \{\sim_i\}_{i \in \mathbf{I}}, \mathfrak{A}, V_N \rangle$, where $W \neq \varnothing$, each $\sim_i$ is an equivalence relation, $\mathfrak{A} = \langle A, \mathfrak{a} \rangle$ is a rigid τ-structure, and $V_N : \mathbf{N} \times W \to A$ assigns values to names at worlds. All worlds of $\mathcal{M}$ share the same structure $\mathfrak{A}$.

The induced term valuation $V_T : \mathrm{Term} \times W \to A$ is:

$$V_T(a, w) = V_N(a, w)$$
$$V_T(c, w) = \mathfrak{a}(c)$$
$$V_T(f(\bar{t}), w) = \mathfrak{a}(f)(V_T(\bar{t}, w)).$$

Semantics

The satisfaction relation is defined as follows:

$\mathcal{M}, w \models t_1 \approx t_2$	$\iff$	$V_T(t_1, w) = V_T(t_2, w)$
$\mathcal{M}, w \models P\bar{t}$	$\iff$	$V_T(\bar{t}, w) \in \mathfrak{a}(P)$
$\mathcal{M}, w \models K_i\varphi$	$\iff$	for every u with $w \sim_i u$, $\mathcal{M}, u \models \varphi$
$\mathcal{M}, w \models Kv_i(\varphi, t)$	$\iff$	for all u, v with $w \sim_i u$ and $w \sim_i v$, if $\mathcal{M}, u \models \varphi$ and $\mathcal{M}, v \models \varphi$, then $V_T(t, u) = V_T(t, v)$
$\mathcal{M}, w \models [\varphi]\psi$	$\iff$	$\mathcal{M}, w \models \varphi$ implies $\mathcal{M}\|_\varphi, w \models \psi$

Here $\mathcal{M}|_\varphi$ restricts $\mathcal{M}$ to the φ-worlds.

Two Semantics

Let T be a consistent universal τ-theory, and put

$$\mathsf{MOD}(T) = \{\mathfrak{B} \mid \mathfrak{B} \models_{\mathrm{FOL}} T\}.$$

Definition (Semantics over $\mathsf{MOD}(T)$)

Write $\Gamma \models_T \varphi$, if for every pointed model $(\mathcal{M}, w)$ based on a first-order structure in $\mathsf{MOD}(T)$, if $\mathcal{M}, w \models \Gamma$, then $\mathcal{M}, w \models \varphi$.

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Definition (Semantics over a fixed structure $\mathfrak{A}$)

Write $\Gamma \models_{\mathfrak{A}} \varphi$, if for every pointed model $(\mathcal{M}, w)$ based on the first-order structure $\mathfrak{A}$, if $\mathcal{M}, w \models \Gamma$, then $\mathcal{M}, w \models \varphi$.

Axiom System $\mathbb{ELKVR}(\mathbb{T})$

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Axiom Schemas

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SI	$t \approx t' \to (\alpha \to \alpha(t'/t))$ where α is an atomic formula
Kv^r4	$Kv_i(\varphi, t) \to K_iKv_i(\varphi, t)$
KVE1	$Kv_it_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$
KVE2	$Kv_it_1 \wedge Kv_i(\varphi, t_2) \to \big(K_i(\varphi \to \neg(t_1 \approx t_2)) \vee K_i(\varphi \to t_1 \approx t_2)\big)$
RIGIDC	$Kv_ic \qquad (c \in \mathsf{C})$
RIGIDF	$\bigwedge_{k=1}^{n} Kv_it_k \to Kv_if(t_1, \ldots, t_n) \qquad (f \in \mathsf{F})$
RIGIDP	$\bigwedge_{k=1}^{n} Kv_it_k \wedge P(t_1, \ldots, t_n) \to K_iP(t_1, \ldots, t_n) \qquad (P \in \mathsf{P})$
$\mathbb{T}$	$\chi(t_1, \ldots, t_n)$, for every $\forall x_1 \cdots \forall x_n\, \chi(x_1, \ldots, x_n) \in T$, where χ is quantifier-free and $t_1, \ldots, t_n$ are arbitrary terms.

Rules

MP $\dfrac{\varphi \quad \varphi \to \psi}{\psi}$

NECK $\dfrac{\vdash \varphi}{\vdash K_i\varphi}$

NAME $\dfrac{\vdash (Kv_ia \wedge a \approx t) \to \chi}{\vdash \chi}$

where a does not occur in t, χ

Intuition for the Axioms: Part 1

Axiom Schemas

TAUT	all instances of propositional tautologies
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5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
Kv^r4	$Kv_i(\varphi, t) \to K_iKv_i(\varphi, t)$

Kv^r5 : $\neg Kv_i(\varphi, t) \to K_i\neg Kv_i(\varphi, t)$ is also provable.

Rules

MP	$\dfrac{\varphi \quad \varphi \to \psi}{\psi}$
NECK	$\dfrac{\vdash \varphi}{\vdash K_i\varphi}$

Intuition for the Axioms: Part 2

Equality Axioms

Axiom	Schema	Intuition
ID	$t \approx t$	Equality is reflexive.
SI	$t \approx t' \to (\alpha \to \alpha(t'/t))$ where α is atomic	Equal terms can replace each other in atomic formulas.

Symmetry, transitivity are derivable.

Intuition for the Axioms: Part 3

Rigid Constants, Functions, and Predicates

Axiom	Schema
RIGIDC	$Kv_i c$
RIGIDF	$\bigwedge_{k=1}^{n} Kv_i t_k \to Kv_i f(t_1, \ldots, t_n)$
RIGIDP	$\bigwedge_{k=1}^{n} Kv_i t_k \wedge P(t_1, \ldots, t_n) \to K_i P(t_1, \ldots, t_n)$

Here $c \in \mathbf{C}$, $f \in \mathbf{F}$, and $P \in \mathbf{P}$.

These axioms express the rigidity of the symbols in τ.

Intuition for the Axioms: Part 4

Knowing Value and Equality

Axiom	Schema
KVE1	$Kv_i t_1 \land K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$
KVE2	$Kv_i t_1 \land Kv_i(\varphi, t_2) \to \big(K_i(\varphi \to \neg(t_1 \approx t_2)) \lor K_i(\varphi \to t_1 \approx t_2)\big)$

Intuition for the Axioms: Part 5

Background Theory $\mathbb{T}$

For every $\forall x_1 \cdots \forall x_n \, \chi(x_1, \ldots, x_n) \in T$, include all instances

$$\chi(t_1, \ldots, t_n),$$

where χ is quantifier-free and $t_1, \ldots, t_n$ are arbitrary terms.

Intuition for the Axioms: Part 5

Background Theory $\mathbb{T}$

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where χ is quantifier-free and $t_1, \ldots, t_n$ are arbitrary terms.

Universal Theory of $(\mathbb{N}, \mathsf{S}, 0)$	Axiom Schema in $\mathbb{T}$
$\forall x \, \neg 0 \approx Sx$	$\neg 0 \approx St$
$\forall x \forall y \, (Sx \approx Sy \to x \approx y)$	$St_1 \approx St_2 \to t_1 \approx t_2$
$\forall x \, \neg x \approx S^n x \quad (n \geq 1)$	$\neg t \approx S^n t \quad (n \geq 1)$

Intuition for the Axioms: Part 6

Rule	Schema
NAME	$\dfrac{\vdash (Kv_i a \wedge a \approx t) \to \chi}{\vdash \chi}$
	where $a \in \mathbf{N}$ does not occur in t or χ.

Intuition for the Axioms: Part 6

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The NAME rule is central to our completeness proof via Henkin construction.

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	where $a \in \mathbf{N}$ does not occur in t or χ.

The NAME rule is central to our completeness proof via Henkin construction.

The key idea is to witness the current value of a non-rigid name using a fresh name a that is logically rigid for agent i.

Strong Completeness over MOD(T)

Theorem (Soundness and strong completeness)

For every consistent universal τ-theory T and every $\Gamma \cup \{\varphi\} \subseteq \mathbf{ELKVR}_\tau$,

$$\Gamma \vdash_{\mathbb{ELKVR}(\mathbb{T})} \varphi \quad \Longleftrightarrow \quad \Gamma \models_T \varphi.$$

Moreover, every consistent set has a model based on a countable structure satisfying T.

Henkin-style Construction

Fix a countably infinite set $\mathbf{N}'$ of new names disjoint from $\mathbf{N}$, and put $\mathbf{N}^+ = \mathbf{N} \sqcup \mathbf{N}'$. Let $\mathcal{L} = \mathbf{ELKVR}_\tau$ be the original language and $\mathcal{L}^+$ the language obtained by replacing $\mathbf{N}$ with $\mathbf{N}^+$. The language $\mathcal{L}$ is countable.

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Proposition

If a set Σ of $\mathcal{L}$-formulas is $\mathcal{L}$-consistent, then it is also $\mathcal{L}^+$-consistent.

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Fix a countably infinite set $\mathbf{N}'$ of new names disjoint from $\mathbf{N}$, and put $\mathbf{N}^+ = \mathbf{N} \sqcup \mathbf{N}'$. Let $\mathcal{L} = \mathbf{ELKVR}_\tau$ be the original language and $\mathcal{L}^+$ the language obtained by replacing $\mathbf{N}$ with $\mathbf{N}^+$. The language $\mathcal{L}$ is countable.

Proposition

If a set Σ of $\mathcal{L}$-formulas is $\mathcal{L}$-consistent, then it is also $\mathcal{L}^+$-consistent.

Note that it may not hold if $\mathcal{L}$ only has finitely many names.

Henkin-style Construction

We want to construct a canonical frame $\mathcal{F}^c = \langle W^c, \{\sim_i\}_{i \in \mathbf{I}} \rangle$ which has the following properties:

- For each world $w \in W^c$, it is a $\mathcal{L}^+$-MCS. And for each term t and agent i, there is a fresh name a such that $Kv_i a \wedge t \approx a \in w$.

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- Extension Lemma: For each $\mathcal{L}$-consistent formula set Σ, it can be extended to a world $w \in W^c$.
- Existence Lemma: If $\widehat{K}_i\varphi \in w$, there is a $v \in W^c$ such that $\varphi \in v$ and $w \sim_i^c v$.

Henkin-style Construction

Suppose $\widehat{K}_i\varphi \in w$, then we want to find a $v \in W^c$ such that $\varphi \in v$ and $w \sim_i^c v$. And we also want v has witness a for each term t and agent j.

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Repeating this process, we can see that w have to contain witnesses for arbitrary finite depth.

Henkin-style Construction

Definition ($\mathcal{L}^+$-saturated set)

A set Γ of $\mathcal{L}^+$-formulas is saturated if it is an $\mathcal{L}^+$-maximal consistent set and satisfies the following condition. For every $k \geq 0$, $\mathcal{L}^+$-formulas $\phi_0, \ldots, \phi_k$, agents $i_1, \ldots, i_k, j \in \mathbf{I}$, and $\mathcal{L}^+$-term t, if

$$\phi_0 \wedge \widehat{K}_{i_1}\big(\phi_1 \wedge \widehat{K}_{i_2}(\phi_2 \cdots \wedge \widehat{K}_{i_k}(\phi_k \wedge \top))\big) \in \Gamma,$$

then there is a name $a \in \mathbf{N}^+$, not occurring in $\phi_0, \ldots, \phi_k, t$, such that

$$\phi_0 \wedge \widehat{K}_{i_1}\big(\phi_1 \wedge \widehat{K}_{i_2}(\phi_2 \cdots \wedge \widehat{K}_{i_k}(\phi_k \wedge (Kv_j a \wedge a \approx t)))\big) \in \Gamma.$$

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For $k = 0$, the two formulas are $\phi_0 \wedge \top$ and $\phi_0 \wedge (Kv_j a \wedge a \approx t)$, respectively.

Rule NAME*

For every $k \geq 0$, if we have the following rule, we can prove the Extension Lemma.

$$\frac{\vdash \phi_0 \wedge \widehat{K}_{i_1}\big(\phi_1 \wedge \cdots \wedge \widehat{K}_{i_k}(\phi_k \wedge (Kv_j a \wedge a \approx t))\cdots\big) \to \bot}{\vdash \phi_0 \wedge \widehat{K}_{i_1}\big(\phi_1 \wedge \cdots \wedge \widehat{K}_{i_k}(\phi_k \wedge \top)\cdots\big) \to \bot} \qquad \text{(NAME*)}$$

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where $a \in \mathbf{N}^+$ does not occur in $\phi_0, \ldots, \phi_k, t$.

For $k = 0$, it is just the NAME rule.

$$\frac{\vdash \phi_0 \wedge (Kv_j a \wedge a \approx t) \to \bot}{\vdash \phi_0 \wedge \top \to \bot}.$$

From NAME to NAME*

In every normal modal proof system containing axiom B,

$$\vdash \varphi \wedge \widehat{K}_i \psi \to \chi \quad \Longleftrightarrow \quad \vdash \psi \to K_i(\varphi \to \chi). \tag{DRB}$$

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For $k = 1$, NAME* has the form

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where $a \in \mathbf{N}^+$ does not occur in ϕ_0, ϕ_1, t.

From NAME to NAME*

We can prove NAME* using NAME and the DRB rule.

(1)	$\vdash \phi_0 \wedge \widehat{K}_{i_1}(\phi_1 \wedge (Kv_j a \wedge a \approx t)) \to \bot$	Premise
(2)	$\vdash \phi_1 \wedge (Kv_j a \wedge a \approx t) \to K_{i_1}(\phi_0 \to \bot)$	1, DRB
(3)	$\vdash (Kv_j a \wedge a \approx t) \to (\phi_1 \to K_{i_1}(\phi_0 \to \bot))$	2, PC
(4)	$\vdash \top \to (\phi_1 \to K_{i_1}(\phi_0 \to \bot))$	3, NAME
(5)	$\vdash \phi_1 \wedge \top \to K_{i_1}(\phi_0 \to \bot)$	4, PC
(6)	$\vdash \phi_0 \wedge \widehat{K}_{i_1}(\phi_1 \wedge \top) \to \bot$	5, DRB.

Here a does not occur in ϕ_0, ϕ_1, t.

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For each $w \in W^c$, if $Kv_i(\varphi, t) \in w$, then there is a name a such that $Kv_i a \wedge K_i(\varphi \to a \approx t) \in w$.

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Proposition

For each $w \in W^c$, if $Kv_i(\varphi, t) \in w$, then there is a name a such that $Kv_i a \wedge K_i(\varphi \to a \approx t) \in w$. if $\neg Kv_i(\varphi, t) \in w$, then there is a name a such that $Kv_i a \wedge \widehat{K}_i(\varphi \wedge a \approx t) \wedge \widehat{K}_i(\varphi \wedge \neg a \approx t) \in w$.

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Therefore, it remains to construct a first-order structure and a valuation such that the Truth Lemma holds for (negated) atomic formulas and formulas of the form $Kv_i a$.

Why Tree Unraveling Necessary

Let $a, b \in \mathbf{N}$ be non-rigid names.

$$\{Kv_1 a, \neg Kv_1 b, Kv_2 b, \neg Kv_2 a, a \approx b\}$$

$$1 \mid 2$$

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We cannot give a valuation! So tree-unraveling is needed. After unraveling, the remaining proof is easy.

The Henkin-style Method

I learnt this method from Corsi and Orlandelli's work on FOIL (2021).

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- The set of names must be infinite.
- It does not work if equality is not included in the language.

Fixed Structure Semantics

Now we have proved is:

Theorem (Soundness and strong completeness)

For every consistent universal τ-theory T and every $\Gamma \cup \{\varphi\} \subseteq \mathbf{ELKVR}_\tau$,

$$\Gamma \vdash_{\mathbb{ELKVR}(\mathbb{T})} \varphi \quad \Longleftrightarrow \quad \Gamma \models_T \varphi.$$

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Recall the fixed structure semantics is:

Definition (Semantics over a fixed structure $\mathfrak{A}$)

Write $\Gamma \models_{\mathfrak{A}} \varphi$, if for every pointed model $(\mathcal{M}, w)$ based on the first-order structure $\mathfrak{A}$, if $\mathcal{M}, w \models \Gamma$, then $\mathcal{M}, w \models \varphi$.

Fixed Structure Semantics

Fix a nonempty τ-structure $\mathfrak{A}$. We assume that, for every quantifier-free τ-formula $\chi(\bar{x})$,

$$\mathfrak{A} \models_{\mathrm{FOL}} \forall \bar{x}\, \chi(\bar{x}) \quad \Longleftrightarrow \quad T \models_{\mathrm{FOL}} \forall \bar{x}\, \chi(\bar{x}). \tag{*}$$

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Then we prove that:

Theorem (Weak Completeness over a Fixed Structure)

Assume (*) holds. Then for every $\varphi \in \mathsf{ELKVR}_\tau$,

$$\vdash_{\mathbb{ELKVR}(\mathbb{T})} \varphi \quad \Longleftrightarrow \quad \models_{\mathfrak{A}} \varphi.$$

Weak Completeness

Given a consistent φ, by the previous theorem, there is a model $\mathcal{M}$ based on a structure $\mathfrak{B} \models_{\mathrm{FOL}} T$ such that $\mathcal{M}, w \models \varphi$.

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We have the finite-world property:

Theorem

If an ELKVR_τ *formula is satisfiable over a structure* $\mathfrak{B}$*, it is satisfiable in a model over the same structure* $\mathfrak{B}$ *with finitely many worlds.*

FMP: Tree Unraveling and Selection

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FMP: Tree Unraveling and Selection

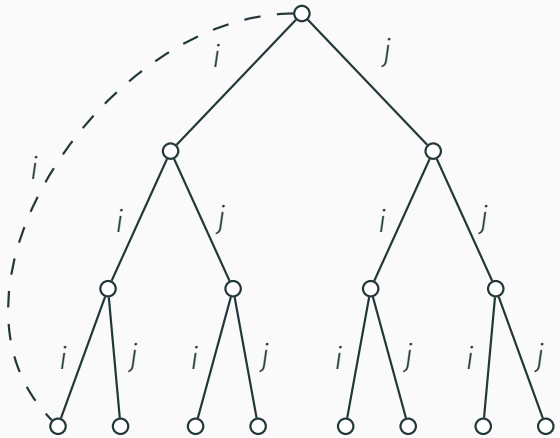
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- If a selected world satisfies $\neg Kv_i(\chi, t)$, select two accessible χ-worlds where t has different values.
- The selected submodel is finite, and truth of the relevant subformulas is preserved by induction on modal depth.

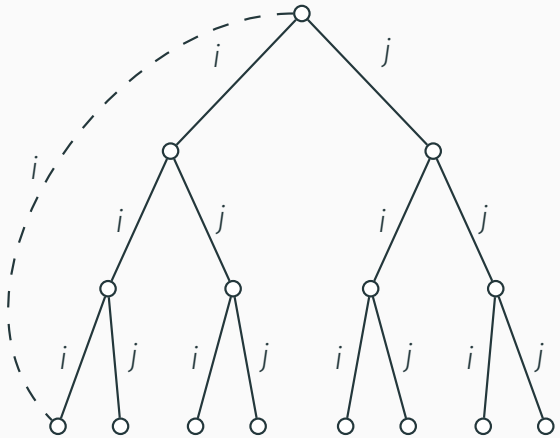
The Selection Trick

Direct selection

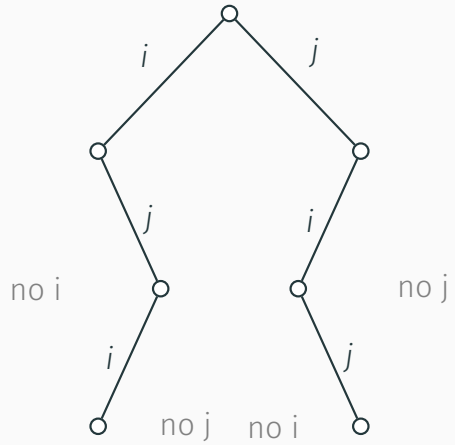


The Selection Trick

Direct selection



Our selection



If we select nodes directly, the hierarchy of depth will collapse after taking the S5 closure.

Weak Completeness over a Fixed Structure

Now we have a model $\mathcal{M}$ based on $\mathfrak{B}$ with only finitely many worlds such that $\mathcal{M}, w \models \varphi$.

Since φ contains only finitely many terms and $\mathcal{M}$ has finitely many worlds, we can record all information relevant to φ in a single quantifier-free first-order formula.

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At each world u, replace names in the terms of φ by distinct first-order variables:

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- For each predicate P and every term t: $P(\bar{t})$ or $\neg P(\bar{t})$.

Weak Completeness: Transfer to $\mathfrak{A}$

Let $\theta(\bar{x})$ be the conjunction of these facts. It is finite and quantifier-free, and

$$\mathfrak{B} \models \exists \bar{x}\, \theta(\bar{x}).$$

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Then all the relevant facts in $\mathcal{M}$ can be realized in a model based on $\mathfrak{A}$, and we can transfer the truth of φ to that model.

Strong Completeness over a Fixed Structure

Definition (Countable quantifier-free diagram realization)

Let $\Delta(z_0, z_1, \ldots)$ be a countable set of quantifier-free τ-formulas. If every finite subset is realizable in $\mathfrak{A}$, then all of Δ is realizable by one valuation in $\mathfrak{A}$.

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3. $\mathfrak{A}$ *has countable quantifier-free diagram realization.*
4. *The fixed-structure semantics* $\models_{\mathfrak{A}}$ *is compact: every finitely satisfiable set of formulas is satisfiable.*

Assignment with Background Theories

Language

Definition (Language RELAS_τ)

Keep $\mathbf{N}$, $\mathbf{I}$, and the rigid signature $\tau = \langle \mathbf{C}, \mathbf{F}, \mathbf{P} \rangle$ as before. Add a countably infinite set $\mathbf{X}$ of variables, disjoint from them.

$$t ::= x \mid a \mid c \mid f(\bar{t}),$$
$$\varphi ::= \top \mid t \approx t \mid P\bar{t} \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid [x := t]\varphi,$$

where $x \in \mathbf{X}$, $a \in \mathbf{N}$, $c \in \mathbf{C}$, $f \in \mathbf{F}$, $P \in \mathbf{P}$, and $i \in \mathbf{I}$. Function and predicate symbols have positive arity.

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$$Kv_i t \quad \text{is expressed by} \quad [x := t]K_i(x \approx t),$$

where x does not occur in t.

Semantics

Definition (Models)

A (constant domain S5) model is a tuple $\mathcal{M} = \langle W, \{\sim_i\}_{i \in \mathsf{I}}, \mathfrak{A}, \eta \rangle$, where $W \neq \varnothing$, each $\sim_i$ is an equivalence relation, $\mathfrak{A} = \langle A, \mathfrak{a} \rangle$ is a rigid τ-structure, and $\eta : \mathbf{N} \times W \to A$ assigns values to names at worlds. All worlds of $\mathcal{M}$ share the same structure $\mathfrak{A}$.

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Definition (Models)

A (constant domain S5) model is a tuple $\mathcal{M} = \langle W, \{\sim_i\}_{i \in \mathsf{I}}, \mathfrak{A}, \eta \rangle$, where $W \neq \varnothing$, each $\sim_i$ is an equivalence relation, $\mathfrak{A} = \langle A, \mathfrak{a} \rangle$ is a rigid τ-structure, and $\eta : \mathbf{N} \times W \to A$ assigns values to names at worlds. All worlds of $\mathcal{M}$ share the same structure $\mathfrak{A}$.

A variable assignment is a map $\sigma : \mathbf{X} \to A$. The induced term valuation $\sigma_w : \mathrm{Term} \to A$ is:

$$\begin{aligned} \sigma_w(x) &= \sigma(x) \\ \sigma_w(a) &= \eta(a, w) \\ \sigma_w(c) &= \mathfrak{a}(c) \\ \sigma_w(f(\bar{t})) &= \mathfrak{a}(f)(\sigma_w(\bar{t})). \end{aligned}$$

Semantics

The satisfaction relation is defined as follows:

$$
\begin{array}{lcl}
\mathcal{M}, w, \sigma \models t_1 \approx t_2 & \iff & \sigma_w(t_1) = \sigma_w(t_2) \\
\mathcal{M}, w, \sigma \models P\overline{t} & \iff & \sigma_w(\overline{t}) \in \mathfrak{a}(P) \\
\mathcal{M}, w, \sigma \models K_i \varphi & \iff & \text{for every } u \text{ with } w \sim_i u, \\
 & & \mathcal{M}, u, \sigma \models \varphi \\
\mathcal{M}, w, \sigma \models [x := t]\varphi & \iff & \mathcal{M}, w, \sigma[x \mapsto \sigma_w(t)] \models \varphi
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Here $\sigma[x \mapsto d]$ maps x to d and agrees with σ on all other variables.

Two Semantics

Let T be a consistent universal τ-theory, and put

$$\mathsf{MOD}(T) = \{\mathfrak{B} \mid \mathfrak{B} \models_{\mathrm{FOL}} T\}.$$

Definition (Semantics over $\mathsf{MOD}(T)$)

Write $\Gamma \models_T \varphi$, if for every pointed model $(\mathcal{M}, w)$ based on a first-order structure in $\mathsf{MOD}(T)$ and every variable assignment σ, if $\mathcal{M}, w, \sigma \models \Gamma$, then $\mathcal{M}, w, \sigma \models \varphi$.

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Definition (Semantics over a fixed structure $\mathfrak{A}$)

Write $\Gamma \models_{\mathfrak{A}} \varphi$, if for every pointed model $(\mathcal{M}, w)$ based on the first-order structure $\mathfrak{A}$ and every variable assignment σ, if $\mathcal{M}, w, \sigma \models \Gamma$, then $\mathcal{M}, w, \sigma \models \varphi$.

Axiom System $\mathbb{S}5\mathbb{RELAS}(\mathbb{T})$

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Axiom Schemas

TAUT	all instances of propositional tautologies
DISTK	$K_i(\varphi \to \psi) \to (K_i\varphi \to K_i\psi)$
T	$K_i\varphi \to \varphi$
4	$K_i\varphi \to K_iK_i\varphi$
5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
ID	$t \approx t$
SI	$t \approx t' \to (\alpha \to \alpha(t'/t))$
	where α is an atomic formula
RIGIDAT	$\alpha \to K_i\alpha$, where α is a name-free atomic formula
ASSUB	$y \approx t \to \big([x := t]\varphi \leftrightarrow \varphi[y/x]\big)$,
	where $\varphi[y/x]$ is admissible (no variable capture)
$\mathbb{T}$	$\chi(y_1, \ldots, y_n)$, for every $\forall x_1 \cdots \forall x_n\, \chi(x_1, \ldots, x_n) \in T$,
	where χ is quantifier-free and $y_1, \ldots, y_n \in \mathbf{X}$.

Rules

MP $\dfrac{\varphi \quad \varphi \to \psi}{\psi}$

NECK $\dfrac{\vdash \varphi}{\vdash K_i\varphi}$

WIT $\dfrac{\vdash (y \approx t) \to \chi}{\vdash \chi}$

where y does not occur in t, χ

Strong Completeness over MOD(T)

Theorem (Soundness and strong completeness)

For every consistent universal τ-theory T and every $\Gamma \cup \{\varphi\} \subseteq \mathbf{RELAS}_\tau$,

$$\Gamma \vdash_{\mathbb{S}5\mathbb{R}\mathbb{E}\mathbb{L}\mathbb{A}\mathbb{S}(\mathbb{T})} \varphi \quad \Longleftrightarrow \quad \Gamma \models_T \varphi.$$

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Moreover, every consistent set has a model based on a countable structure satisfying T.

Weak Completeness

The weak completeness over a fixed structure may fail.

Consider the successor arithmetic. For the countably saturated model $\mathfrak{N}^* = (\mathbb{N}^*, \mathsf{S}^*, 0)$, we still have

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However, consider the following formula:

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It is satisfiable over $\mathfrak{N}^*$, but not over $\mathfrak{N}$.

Non-Recursive Enumerability

Proposition

For $\mathfrak{N} = (\mathbb{N}, \mathsf{S}, 0)$, the set of formulas valid over $\mathfrak{N}$ is not recursively enumerable.

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Theorem

Assume $(*)$ and that T is recursively enumerable. If $\mathbb{S}5\mathbb{RELAS}(\mathbb{T})$ is not weakly complete over $\mathfrak{A}$, then

$$\{\varphi \in \mathsf{RELAS}_\tau \mid\mid\!\!=_{\mathfrak{A}} \varphi\}$$

is not recursively enumerable.

Thanks!